

SIMATS ENGINEERING
Co3-ASSESSMENT TOOL 2
Case study

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Dr.V.Parthipan

QUESTIONS:4

Total Marks:40

Code of Conduct:

*I **ANTONY MAGRIN (192572103)** my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Criteria	Understanding the case 2.5	Application of Concepts 2.5	Analysis 2	Solution Design 2	Clarity 1	Total (10)
Q1						
Q2						
Q3						
Q4						

Faculty Signature

QUESTIONS:4

Total Marks:40

1. A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss, 180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2. A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS, DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

3. A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

4. Examine the impact of centralized DNS architecture on application performance.
5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations**.
2. Explain how each factor affects transaction processing in high-frequency systems
3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

ANSWERS

Q1)

1. Analyze whether the problem is primarily caused by the transport protocol (TCP/UDP) or the application layer.

The problem is mainly caused by the use of the **UDP transport protocol** together with an **inefficient application-layer buffering strategy**. Real-time video conferencing generally uses UDP because it provides low latency and does not retransmit lost packets. However, an **8% packet loss, 180 ms jitter**, and fluctuating bandwidth cause voice distortion, frozen video, and communication delays. The application layer also contributes if its jitter buffer is not properly configured to compensate for network variations.

2. Justify your answer based on the communication requirements of real-time multimedia applications.

Real-time multimedia applications require:

- Low latency for smooth communication.
- Minimal jitter for synchronized audio and video.
- Continuous packet delivery instead of perfect reliability.
- Fast recovery from network congestion.

UDP is preferred because retransmissions in TCP introduce additional delay, making conversations unnatural. Therefore, conferencing applications prioritize speed over guaranteed delivery.

3. Recommend suitable protocol-level or application-level improvements.

- Implement adaptive jitter buffering to smooth packet arrival.
- Use RTP with RTCP for media transmission and quality monitoring.
- Enable Forward Error Correction (FEC) to recover lost packets.
- Apply QoS (Quality of Service) to prioritize voice and video traffic.
- Use adaptive bitrate streaming based on available bandwidth.
- Monitor network congestion continuously and optimize routing.

Q2)

1. Analyze the reasons behind the high DNS resolution time.

The DNS lookup time of **800 ms** may be caused by:

- Centralized DNS servers located far from users.
- Network congestion and high latency.
- Poor DNS caching configuration.
- High TTL expiration causing frequent DNS queries.
- Overloaded DNS servers handling excessive requests.

2. Propose a suitable DNS architecture to reduce lookup time below 50 ms.

A suitable architecture includes:

- **Anycast DNS:** Routes users to the nearest DNS server.
- **DNS Pre-fetching:** Resolves frequently visited domains before users request them.
- **TTL Optimization:** Uses balanced TTL values to improve cache efficiency.
- **Distributed DNS Servers:** Deploy servers across multiple geographic regions.
- **Redundant Secondary DNS Servers:** Ensure service continuity during failures.

3. Evaluate the advantages and limitations.

Advantages

- Faster DNS lookup (<50 ms).
- Improved scalability.
- High availability and fault tolerance.
- Reduced server workload.
- Better global performance.

Limitations

- Higher deployment cost.
- More complex configuration.
- Cache synchronization challenges.
- Requires continuous monitoring and maintenance.

Q3)

1. Examine the impact of centralized DNS architecture on application performance.

A centralized DNS architecture negatively affects application performance because:

- All DNS requests are sent to one server.
- Lookup latency increases significantly.
- Server overload causes slow responses.
- A single server failure affects all users.
- User experience deteriorates during peak traffic.

2. Design a distributed DNS solution.

The proposed distributed DNS architecture includes:

- Multiple DNS servers located in different regions.
- Anycast routing to direct users to the nearest DNS server.
- Secondary DNS servers for redundancy.

- Intelligent DNS load balancing.
- Regional caching servers to reduce repeated queries.
- Automatic failover when a DNS server becomes unavailable.

3. Analyze the influence of TTL values, Anycast routing, and DNS caching.

TTL Values

- Higher TTL reduces DNS queries and server load.
- Lower TTL allows faster updates but increases lookup requests.

Anycast Routing

- Directs users to the nearest DNS server.
- Reduces latency and improves response time.
- Provides automatic failover.

DNS Caching

- Stores previously resolved records.
- Reduces lookup delay.
- Minimizes DNS server workload.
- Improves overall application performance.

Q4)

1. Analyze three TCP-related factors responsible for increased latency.

a) Flow Control

The receiver advertises a smaller window size during heavy traffic, forcing the sender to slow transmission and increasing acknowledgment delay.

b) Nagle's Algorithm

Nagle's algorithm combines multiple small packets before sending them. Although it reduces network overhead, it introduces additional delay, which is unsuitable for stock trading systems.

c) SYN Queue Limitations

During peak traffic, the SYN queue becomes full, delaying new TCP connection establishment and increasing response time.

2. Explain how each factor affects high-frequency transaction processing.

- **Flow Control:** Delays order acknowledgments and slows transaction processing.
- **Nagle's Algorithm:** Introduces transmission delay for small trading messages.
- **SYN Queue Overflow:** Prevents new client connections, increasing customer wait

time.

These delays reduce trading efficiency and negatively impact customer satisfaction.

3. Recommend TCP configuration changes.

- Disable Nagle's algorithm using **TCP_NODELAY**.
- Increase TCP receive and send buffer sizes.
- Increase SYN backlog queue size.
- Enable TCP Fast Open to reduce connection setup delay.
- Optimize TCP window scaling.
- Use persistent TCP connections instead of creating new connections repeatedly.
- Continuously monitor retransmissions and network congestion to maintain reliability.